



CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

UNIT 1: ECOLOGY, PEOPLE AND NATURAL RESOURCE USE

PAPER 02

2 hours 30 minutes

06 MAY 2008 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. **DO NOT** open this examination paper until instructed to do so.
2. This paper consists of **NINE** questions, **THREE** from each Module.
3. Candidates **MUST** answer **SIX** questions, **TWO** from **EACH** Module.
4. Write your answers in the answer booklet provided.
5. The use of a silent, non-programmable, scientific calculator is permitted.

MODULE 1

Answer ANY TWO questions.

1. Non-native species are often introduced into stable communities. Through evolution, species develop characteristics that allow them to adapt to conditions in particular environments.

- (a) Name ONE non-native species that was introduced in the Caribbean. [1 mark]
- (b) Explain why non-native species may be successful in new environments. Include TWO reasons in your response. [6 marks]
- (c) Outline THREE impacts on the ecosystem that may result from the introduction of a non-native species. [9 marks]
- (d) Explain the role of natural selection in the adaptation of species to their natural environment. (Include TWO points in your response.) [4 marks]

Total 20 marks

2. Table 1 presents the stability indices for three ecosystems based on the species diversity of each ecosystem.

TABLE 1: ECOSYSTEM STABILITY INDEX

Ecosystem	Ecosystem Stability Index
A	7
B	4
C	10

- (a) Define EACH of the following:
 - (i) Ecosystem [3 marks]
 - (ii) Ecosystem stability [3 marks]
 - (iii) Species diversity [3 marks]

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- (b) (i) Which ecosystem in Table 1 has the GREATEST species diversity? [1 mark]
- (ii) Give ONE reason to support your answer. [2 marks]
- (c) Explain how species diversity can influence ecosystem stability. (Include TWO ways in your response.) [8 marks]

Total 20 marks

3. (a) Figure 1 illustrates the cycling of matter through an ecosystem.

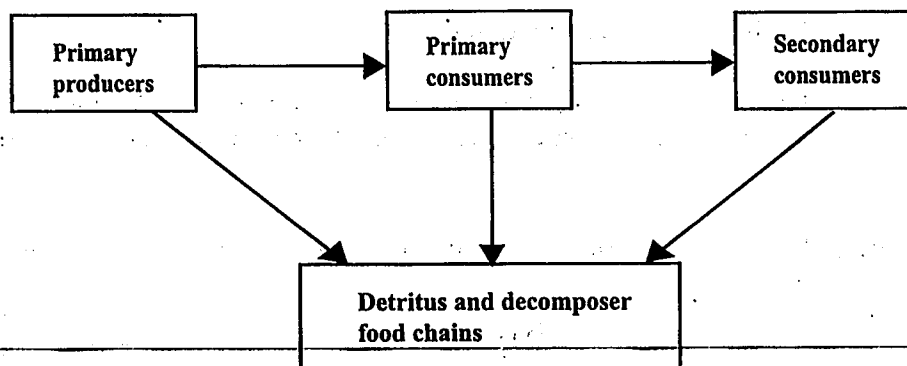


Figure 1. Cycling of matter through an ecosystem

- (i) Name TWO decomposers and state ONE source of energy for EACH. [4 marks]
- (ii) Explain the importance of the decomposer food chain in the cycling of matter illustrated in Figure 1. [4 marks]
- (b) (i) State TWO benefits of natural ecosystems to humans. Give an example of EACH. [4 marks]
- (ii) Describe TWO ways in which human activities can disrupt the integrity of natural ecosystems. [8 marks]

Total 20 marks

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MODULE 2

Answer ANY TWO questions.

4. (a) What is meant by 'sustainable development'? [3 marks]
- (b) Explain how high fertility rates impact on a country's ability to achieve sustainable development. Include an appropriate example in your response. [6 marks]
- (c) (i) Based on your response to 4 (b), suggest an appropriate fertility rate that Caribbean countries should aim at achieving. [1 mark]
- (ii) Justify your choice of fertility rate. [4 marks]
- (d) (i) Describe ONE method or measure of population control that will be effective in assisting your country to achieve the target fertility rate you suggested. [4 marks]
- (ii) Outline the reason why the method you described in 4 (d) (i) will be effective. [2 marks]

Total 20 marks

5. Table 2 presents data on world population numbers in 1990 and the estimated numbers in 2020.

TABLE 2: WORLD POPULATION SIZE

	More Developed Countries (millions)	Less Developed Countries (millions)
Population in 1990	1247	4486
Estimated Population in 2020	1375	6548

- (a) Calculate the estimated percentage growth in world population attributable to Less Developed Countries between 1990 and 2020. [5 marks]
- (b) Explain why this estimated growth in the population of Less Developed Countries should be a cause for concern. [5 marks]

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- (c) Describe ONE environmental impact associated with the trend in world population growth calculated in 5 (a). [4 marks]
- (d) Suggest an approach to mitigate the environmental impact described in 5 (c). [6 marks]

Total 20 marks

6. (a) Explain how EACH of the following impacts on the rate of growth of a population:
- (i) Culture [4 marks]
- (ii) Religion [4 marks]
- (b) Table 3 shows the Human Development Index (HDI) of two countries for 2007.

TABLE 3: HUMAN DEVELOPMENT INDEX

Country	HDI
A	0.89
B	0.38

What THREE deductions can be made regarding the relative achievement of BOTH countries based on their HDI presented in Table 3? [6 marks]

- (c) Evaluate the statement below. (Include at least SIX points in your response.)

The education of women is critical to lowering the population growth rate of Less Developed Countries. [6 marks]

Total 20 marks

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MODULE 3

Answer ANY TWO questions.

7. (a) Describe TWO functions of coral reef ecosystems in the Caribbean. [6 marks]
- (b) Explain how ANY TWO human activities impact on coral reef ecosystems in the Caribbean. [8 marks]
- (c) Describe TWO methods for conserving coral reef ecosystems in the Caribbean. [6 marks]

Total 20 marks

8. Figure 2 shows how the quantities of two natural resources, A and B, change over a ten-year period.

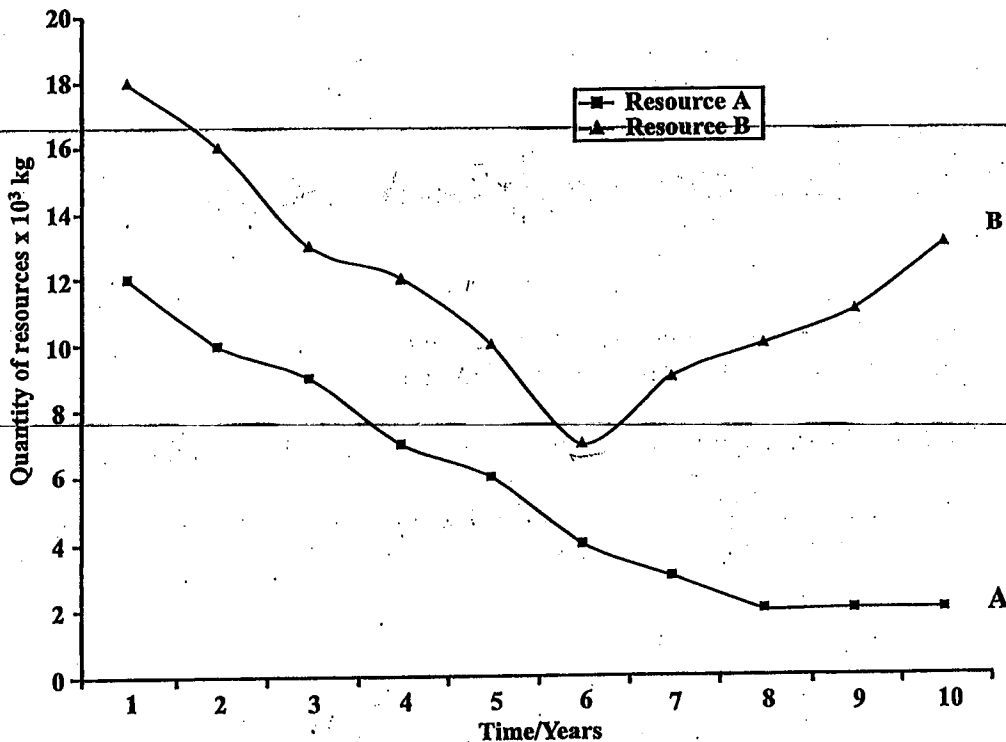


Figure 2. The quantities of two natural resources, A and B, over a ten-year period

- (a) (i) From the data illustrated in Figure 2, identify the resource which is non-renewable. [1 mark]
- (ii) Justify your response to 8 (a) (i). [7 marks]

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- (b) Using suitable examples, distinguish between 'consumptive' and 'non-consumptive' use of natural resources. [6 marks]
- (c) Explain why bioprospecting is considered a non-consumptive use of natural resources. (Include THREE reasons in your response.) [6 marks]

Total 20 marks

- 9.
- (a) Identify TWO categories of protected areas as classified by the International Union for the Conservation of Nature (IUCN). [2 marks]
 - (b) For EACH category of protected area identified in (a), state
 - (i) the purpose [2 marks]
 - (ii) the main feature. [2 marks]
 - (c) Explain how protected areas promote natural resource conservation. Include THREE roles in your response. [6 marks]
 - (d) Assess the effectiveness of protected areas in conserving natural resources in the Caribbean. ~~Include THREE issues associated with the implementation and operation of protected areas.~~ [8 marks]

Total 20 marks

END OF TEST